MLCC 2026

Lab 4/5 – Nonlinear models

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Nonlinear models

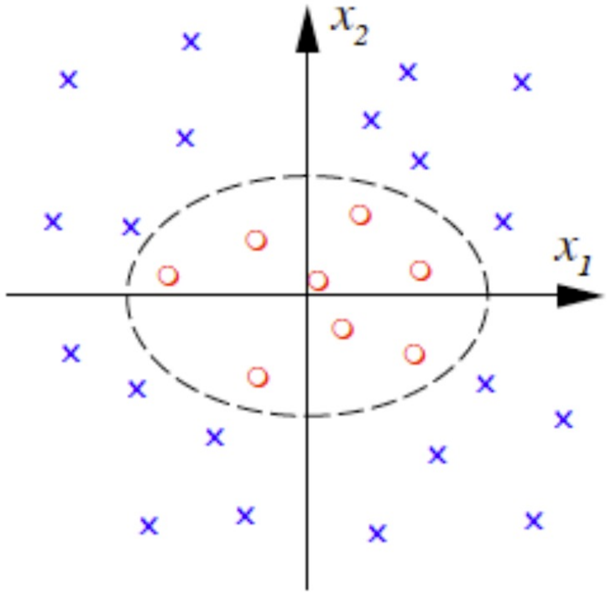
Hypothesis space $\mathcal{H}$ = representation/parametrization of the functions

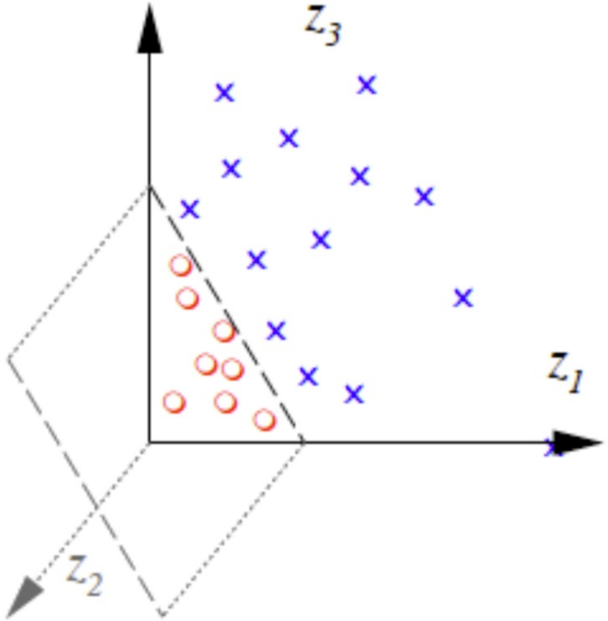
$$\underbrace{w^T x}_{\text{linear}} \qquad \underbrace{w^T \Phi(x) \qquad w^T \sigma(W^T x)}_{\text{nonlinear}}$$

Nonlinear models

$$w^T \Phi(x)$$

$$\Phi : R^2 \to R^3$$
$$(x_1, x_2) \mapsto (z_1, z_2, z_3) := (x_1^2, \sqrt{2}x_1x_2, x_2^2)$$





Nonlinear models

Kernels

$$f(x) = \mathrm{w}^{\mathrm{T}}\Phi(x) \quad \rightarrow \quad f(x) = \sum_{i=1}^{n} c_i \, k(x_i, x) \qquad \left(k(x,x') = \Phi^{\mathrm{T}}(x)\Phi(x')\right)$$

Examples:

- Linear $\qquad k(x,x') = x^T x'$
- Polynomial $\qquad k(x,x') = (x^T x' + 1)^d$
- Gaussian $\qquad k(x,x') = \exp\left(-\frac{\|x-x'\|^2}{2\sigma^2}\right)$

Nonlinear models

How to find c_i:

$$\widehat{K}_{ij} = k(x_i, x_j), \qquad i, j = 1, \ldots, n$$

kernel_matrix

$$(\widehat{K} + n\lambda I)c = \widehat{Y}$$

$$\hat{c} = (\widehat{K} + n\lambda I)^{-1}\,\widehat{Y}$$

krls_train

$$f(x) = \sum_{i=1}^{n} c_i\, k(x_i, x)$$

krls_predict

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Kernels

- Generate data for a nonlinear problem
- Try a linear model
- Try a feature map adapted to the data
- Try a Gaussian kernel